

THE SHARP ONE-DIMENSIONAL LIEB–THIRRING INEQUALITY FOR THE SUM OF EIGENVALUES

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ABSTRACT. We prove that the best constant in the one-dimensional Lieb–Thirring inequality with exponent one is $4/(3\sqrt{3}\pi)$, confirming the Lieb–Thirring conjecture in this case. Moreover, we extend the inequality to operator-valued potentials and obtain the bound $L_{1,d} \leq (2/\sqrt{3})L_{1,d}^{\text{cl}}$ in every dimension.

1. INTRODUCTION

For a one-dimensional Schrödinger operator $-d^2/dx^2 - V$, the lowest negative eigenvalue λ_1 satisfies the sharp bound [Kel61, LT76]

$$|\lambda_1| \leq \frac{4}{3\sqrt{3}\pi} \int_{\mathbb{R}} V(x)_+^{3/2} dx,$$

where $V_+ = \max\{V, 0\}$. Lieb and Thirring [LT76] conjectured that the same estimate holds with $|\lambda_1|$ replaced by the sum of the absolute values of all negative eigenvalues. In this paper we prove this conjecture.

Theorem 1. *Let $V \in L^{3/2}(\mathbb{R})$ be real-valued. Then the negative eigenvalues $\{\lambda_k\}_{k \geq 1}$ of $-d^2/dx^2 - V$ satisfy*

$$\sum_{k \geq 1} |\lambda_k| \leq \frac{4}{3\sqrt{3}\pi} \int_{\mathbb{R}} V(x)_+^{3/2} dx.$$

The constant is sharp.

Equality holds for $V(x) = \frac{3}{4} \text{sech}^2 x$, which produces a single negative eigenvalue, $\lambda_1 = -1/4$.

This question belongs to the general problem of determining the optimal constants in the celebrated Lieb–Thirring inequalities

$$\sum_{k \geq 1} |\lambda_k|^\gamma \leq L_{\gamma,d} \int_{\mathbb{R}^d} V(x)_+^{\gamma+d/2} dx,$$

where $\{\lambda_k\}_{k \geq 1}$ are the negative eigenvalues of $-\Delta - V$ in $L^2(\mathbb{R}^d)$, counted with multiplicity, and $L_{\gamma,d}$ denotes the optimal constant. For real-valued $V \in L^{\gamma+d/2}(\mathbb{R}^d)$, Lieb and Thirring [LT76] proved these bounds for $\gamma > 1/2$ if $d = 1$ and $\gamma > 0$ if $d \geq 2$. The endpoints $\gamma = 1/2$, $d = 1$, and $\gamma = 0$, $d \geq 3$, were established by Weidl [Wei96] and Cwikel, Lieb, and Rozenblum [Cwi77, Lie76, Roz72, Roz76], respectively.

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The lowest eigenvalue and the semiclassical limit give two competing lower bounds on $L_{\gamma,d}$. The first is the sharp constant $L_{\gamma,d}^{(1)}$ for the lowest eigenvalue alone; the opening inequality gives $L_{1,1}^{(1)} = 4/(3\sqrt{3}\pi)$. The second is the semiclassical constant

$$L_{\gamma,d}^{\text{cl}} = \frac{\Gamma(\gamma + 1)}{(4\pi)^{d/2}\Gamma(\gamma + 1 + d/2)},$$

which follows from Weyl asymptotics in the strong-coupling limit. The original conjecture of Lieb and Thirring proposes that the optimal constant is determined by one of these two regimes, namely a single bound state, or the semiclassical limit in which the number of bound states tends to infinity. In terms of the constants, it asserts that

$$L_{\gamma,d} = \max \{L_{\gamma,d}^{(1)}, L_{\gamma,d}^{\text{cl}}\}.$$

Until now, the conjecture had been proved only for $\gamma \geq 3/2$ in every dimension and for $\gamma = 1/2$ in one dimension. The one-dimensional case $\gamma = 3/2$ predates Lieb and Thirring's work and follows from the KdV trace identities [BF60, ZF71, GGKM74]; Aizenman–Lieb monotonicity [AL78] gives all $\gamma > 3/2$. Laptev and Weidl [LW00] established the result for $d > 1$, while Hundertmark, Lieb, and Thomas [HLT98] proved $L_{1/2,1} = L_{1/2,1}^{(1)} = 1/2$. In one dimension, the single-eigenvalue constant is the larger candidate for $1/2 \leq \gamma < 3/2$, and the two constants coincide at $\gamma = 3/2$. Numerical investigations support the one-dimensional conjecture [Lev14] and suggest periodic optimizers in two dimensions [FGL21b].

By contrast, the conjecture fails in several higher-dimensional ranges, including $d = 3$, $1/2 < \gamma < 1$ [HR90, FGL21a]. We refer to the surveys [Fra21, Sch22, Fra23, Nam23] and the monograph [FLW22] for further background.

The exponent $\gamma = 1$ is of particular importance. The dual kinetic inequality is the key estimate in Lieb and Thirring's proof of the stability of matter [LT75]. In one dimension, Eden and Foias [EF91] obtained $L_{1,1} \leq 2/(3\sqrt{3})$, and Dolbeault, Laptev, and Loss [DLL08] extended their argument to matrix-valued potentials. Building on Rumin's direct kinetic method [Rum11], Frank, Hundertmark, Jex, and Nam [FHJN21] developed a method for improving upper bounds on the Lieb–Thirring constants. Their work, together with Corso and Ried's solution of the associated variational problem [CR25], yielded the best previously known dimension-independent bound,

$$L_{1,d} \leq 1.44655 L_{1,d}^{\text{cl}}, \quad d \geq 1.$$

In every dimension, Nam [Nam18] obtained a kinetic bound with constant arbitrarily close to the semiclassical value and a gradient error term. Seiringer and Solovej [SS23] subsequently strengthened this estimate and substantially simplified the proof.

Benguria and Loss [BL04] connected the two-eigenvalue problem with the so-called ovals conjecture, a spectral isoperimetric problem for closed plane curves. This conjecture has been studied in [BT05, Lin06, Den15, BM15, Lin25]. Our proof extends the one-eigenvalue argument of Benguria and Loss, based on a change to cumulative

mass coordinates, to orthonormal families of arbitrary finite size, retaining the sharp constant.

We further extend the kinetic inequality to vector-valued orthonormal families. By duality [DLL08], this gives the one-dimensional Lieb–Thirring inequality for operator-valued potentials, with the same sharp constant. For $d \geq 2$, we apply this estimate in one coordinate and the sharp $\gamma = 3/2$ inequality of Laptev and Weidl [LW00] in the remaining coordinates. Together with Theorem 1, this yields the following bound in every dimension.

Theorem 2. *For every $d \geq 1$, we have*

$$L_{1,d} \leq \frac{2}{\sqrt{3}} L_{1,d}^{\text{cl}}.$$

The factor $2/\sqrt{3} \approx 1.1547$ improves the previous dimension-independent factor 1.44655 obtained through the contributions of Frank, Hundertmark, Jex, and Nam [FHJN21] and Corso and Ried [CR25].

Section 2 proves Theorem 1 by extending the cumulative mass argument to finite orthonormal families. Section 3 establishes the vector-valued extension and derives Theorem 2 by dimension lifting.

2. PROOF OF THEOREM 1

In this section, we work with real-valued functions. By the standard duality between the spectral and kinetic Lieb–Thirring inequalities [Sch22, Section 2], Theorem 1 follows from the following proposition.

Proposition 3. *Let N be a positive integer and let $u_1, \dots, u_N \in H^1(\mathbb{R})$ be real-valued and orthonormal in $L^2(\mathbb{R})$. Then*

$$\sum_{j=1}^N \int_{\mathbb{R}} (u'_j)^2 dx \geq \frac{\pi^2}{4} \int_{\mathbb{R}} \left(\sum_{j=1}^N u_j^2 \right)^3 dx. \quad (2.1)$$

We first derive Theorem 1 from this proposition. We then recall the cumulative mass argument of Benguria and Loss and extend its local form to orthonormal families, proving the proposition at the end of the section.

Proof of Theorem 1. Applying Proposition 3 to finite orthonormal families of real eigenfunctions corresponding to negative eigenvalues gives the stated bound by the standard duality. Sharpness follows from the equality case stated in the introduction. $\square$

The scalar case. For $N = 1$, fix a real $u \in H^1(\mathbb{R})$ with $\|u\|_2 = 1$. The bound is the sharp inequality of Sz.-Nagy [SzN41],

$$\int_{\mathbb{R}} (u')^2 dx \geq \frac{\pi^2}{4} \int_{\mathbb{R}} u^6 dx. \quad (2.2)$$

Benguria and Loss [BL04, Section 2] give a proof which reduces this inequality to a Dirichlet Poincaré inequality on $(0, 1)$. Introduce the cumulative mass coordinate

$$s = a(x) = \int_{-\infty}^x u(y)^2 dy, \quad w(s) = u(x)^2.$$

By normalisation,

$$a(-\infty) = 0, \quad a(+\infty) = 1, \quad a' = u^2, \quad 0 \leq a \leq 1.$$

In these variables,

$$\int_{\mathbb{R}} (u')^2 dx = \frac{1}{4} \int_0^1 \dot{w}^2 ds, \quad \int_{\mathbb{R}} u^6 dx = \int_0^1 w^2 ds, \quad w(0) = w(1) = 0.$$

Thus the Dirichlet Poincaré inequality with constant π^2 gives (2.2); see [FLW22, Section 2.5].

We use a pointwise form of the Poincaré inequality with a derivative correction that vanishes after integration. Fix $\alpha \in (0, \pi)$ and suppose that $q_\alpha \in C^1([0, 1]; \mathbb{R})$ satisfies

$$\dot{q}_\alpha + q_\alpha^2 \leq -\alpha^2. \quad (2.3)$$

Then, for real $w \in H_0^1(0, 1)$,

$$\dot{w}^2 - \frac{d}{ds}(q_\alpha w^2) = (\dot{w} - q_\alpha w)^2 - (\dot{q}_\alpha + q_\alpha^2)w^2 \geq \alpha^2 w^2. \quad (2.4)$$

The correction integrates to zero because $w(0) = w(1) = 0$ and q_α is bounded. Hence

$$\int_0^1 \dot{w}^2 ds \geq \alpha^2 \int_0^1 w^2 ds. \quad (2.5)$$

We choose, here and throughout the remainder of the paper,

$$q_\alpha(s) = -\alpha \tan(\alpha(s - 1/2)), \quad 0 < \alpha < \pi. \quad (2.6)$$

This satisfies the condition (2.3) with equality. Letting $\alpha \uparrow \pi$ in (2.5) proves Poincaré's inequality with constant π^2 .

Remark 4. As $\alpha \uparrow \pi$, $q_\alpha(s)$ converges on $(0, 1)$ to $\pi \cot(\pi s)$, the logarithmic derivative of the first Dirichlet eigenfunction $\sin(\pi \cdot)$. The approximation by bounded functions avoids an additional endpoint argument.

Pulling (2.4) back through the change of variables of Benguria and Loss, the local correction is

$$\frac{u^2}{4} \frac{d}{ds}(q_\alpha w^2) \Big|_{s=a(x)} = \frac{1}{4} \frac{d}{dx}(u^4 q_\alpha(a)).$$

This gives, almost everywhere,

$$(u')^2 - f'_\alpha \geq \frac{\alpha^2}{4} u^6, \quad f_\alpha = \frac{1}{4} u^4 q_\alpha(a). \quad (2.7)$$

Since u tends to zero at both ends of the line and q_α is bounded, $u^4 q_\alpha(a)$ vanishes there. Integrating and letting $\alpha \uparrow \pi$ proves (2.2).

Orthonormal families. Write $U = (u_1, \dots, u_N)^\top$ and define the cumulative mass matrix

$$A(x) = \int_{-\infty}^x W(y) dy, \quad W = UU^\top.$$

By orthonormality,

$$A(-\infty) = 0, \quad A(+\infty) = I, \quad A' = W.$$

For every unit vector c , the quantity $c \cdot A(x)c$ is the L^2 mass of the normalised function $c \cdot U$ on $(-\infty, x)$. Hence

$$0 \leq A \leq I.$$

Generalising (2.7), we seek for any $\alpha \in (0, \pi)$ a correction F_α satisfying

$$|U'|^2 - F'_\alpha \geq \frac{\alpha^2}{4}|U|^6 \quad \text{a.e. on } \mathbb{R}, \quad \lim_{x \rightarrow \pm\infty} F_\alpha(x) = 0. \quad (2.8)$$

The scalar correction is quadratic in u^2 , so we seek a correction quadratic in W . Equip $\mathbb{R}^{N \times N}$ with the Hilbert–Schmidt inner product $\langle B, C \rangle_{\text{HS}} = \text{Tr}(B^\top C)$. To let A act symmetrically on the two indices of W , introduce

$$\mathcal{M}_A(B) = \frac{1}{2}(AB + BA).$$

This operator is self-adjoint and preserves symmetric matrices. If $Ae_i = \mu_i e_i$ in an orthonormal basis, then $e_i e_j^\top$ form an orthonormal eigenbasis of $\mathcal{M}_A$, with eigenvalues $(\mu_i + \mu_j)/2$. Its spectrum lies in $[0, 1]$, so the spectral theorem defines $q_\alpha(\mathcal{M}_A)$. We set

$$F_\alpha = \frac{1}{4} \langle W, q_\alpha(\mathcal{M}_A) W \rangle_{\text{HS}}. \quad (2.9)$$

For $N = 1$, this agrees with f_α . We verify the pointwise bound by differentiating F_α , completing the square, and using a two-point identity for q_α .

In the proof of Proposition 3, we use the following Daleckii–Krein formula [DK51]; see also Bhatia [Bha07, Theorem 5.3.1].

Lemma 5 (Daleckii–Krein). *Let $f \in C^1(\mathbb{R}; \mathbb{R})$. The map $T \mapsto f(T)$ on the self-adjoint operators on a finite-dimensional Hilbert space is continuously differentiable. For any self-adjoint H ,*

$$\text{D}f(T)[H] = \sum_{\mu, \nu \in \text{spec}(T)} \frac{f(\mu) - f(\nu)}{\mu - \nu} P_\mu H P_\nu.$$

Here P_μ is the spectral projection associated to μ , and the quotient is interpreted as $f'(\mu)$ when $\mu = \nu$.

Proof of Proposition 3. Fix $\alpha \in (0, \pi)$ and retain the definitions above. Since $U \in H^1(\mathbb{R}; \mathbb{R}^N)$, the matrix W is continuous and locally absolutely continuous, and $A \in C^1$ with $A' = W$. After extending q_α from $[0, 1]$ to a C^1 function on $\mathbb{R}$, Lemma 5 shows

that F_α is locally absolutely continuous. The following differentiations are therefore valid almost everywhere.

Set $B = q_\alpha(\mathcal{M}_A)W$. The product rule and self-adjointness of $q_\alpha(\mathcal{M}_A)$ give

$$F'_\alpha = \frac{1}{2}\langle W', B \rangle_{\text{HS}} + \frac{1}{4}\left\langle W, \left(\frac{d}{dx}q_\alpha(\mathcal{M}_A)\right)W \right\rangle_{\text{HS}}.$$

Since $W' = U'U^\top + U(U')^\top$ and B is symmetric,

$$\frac{1}{2}\langle W', B \rangle_{\text{HS}} = \frac{1}{2}(U' \cdot BU + U \cdot BU') = U' \cdot BU.$$

Thus

$$F'_\alpha = U' \cdot BU + \frac{1}{4}\left\langle W, \left(\frac{d}{dx}q_\alpha(\mathcal{M}_A)\right)W \right\rangle_{\text{HS}}.$$

To prove the pointwise inequality in (2.8), we complete the square:

$$\begin{aligned} |U'|^2 - F'_\alpha &= \left|U' - \frac{1}{2}BU\right|^2 \\ &\quad - \frac{1}{4}\left(|BU|^2 + \left\langle W, \left(\frac{d}{dx}q_\alpha(\mathcal{M}_A)\right)W \right\rangle_{\text{HS}}\right). \end{aligned} \tag{2.10}$$

It remains to control the expression in brackets.

At a fixed point $x \in \mathbb{R}$, choose an orthonormal eigenbasis $Ae_i = \mu_i e_i$ and put $s_{ij} = (\mu_i + \mu_j)/2$. The matrices $E_{ij} = e_i e_j^\top$ form an orthonormal eigenbasis of $\mathcal{M}_A$ with eigenvalues s_{ij} , and $W_{ij} = (e_i \cdot U)(e_j \cdot U)$. Thus

$$q_\alpha(\mathcal{M}_A)E_{ij} = q_\alpha(s_{ij})E_{ij}, \quad B = \sum_{i,j} q_\alpha(s_{ij})W_{ij}E_{ij}.$$

In particular,

$$B_{ij} = q_\alpha(s_{ij})W_{ij}.$$

We first expand the square term,

$$|BU|^2 = \text{Tr}(BW B) = \sum_{i,j,k} B_{ij}W_{jk}B_{ki} = \sum_{i,j,k} W_{ij}W_{jk}W_{ki}q_\alpha(s_{ij})q_\alpha(s_{ki}).$$

Write $q_\alpha[s, t] = (q_\alpha(s) - q_\alpha(t))/(s - t)$, with $q_\alpha[s, s] = \dot{q}_\alpha(s)$. We apply Lemma 5 with $T = \mathcal{M}_A$ and $H = \mathcal{M}_W$. Since

$$\mathcal{M}_W(E_{ij}) = \frac{1}{2}\sum_k (W_{ki}E_{kj} + W_{jk}E_{ik}),$$

the formula gives

$$Dq_\alpha(\mathcal{M}_A)[\mathcal{M}_W]E_{ij} = \frac{1}{2}\sum_k (q_\alpha[s_{kj}, s_{ij}]W_{ki}E_{kj} + q_\alpha[s_{ik}, s_{ij}]W_{jk}E_{ik}).$$

Using $W = \sum_{i,j} W_{ij} E_{ij}$ and pairing with W , we obtain

$$\begin{aligned} \left\langle W, \left(\frac{d}{dx} q_\alpha(\mathcal{M}_A) \right) W \right\rangle_{\text{HS}} &= \frac{1}{2} \sum_{i,j,k} W_{ij} W_{jk} W_{ki} (q_\alpha[s_{kj}, s_{ij}] + q_\alpha[s_{ik}, s_{ij}]) \\ &= \sum_{i,j,k} W_{ij} W_{jk} W_{ki} q_\alpha[s_{ij}, s_{ki}]. \end{aligned}$$

Indeed, interchanging i and j in the first sum makes the two contributions equal, using symmetry of W and of the divided difference. Substituting into (2.10) and discarding the non-negative square, we obtain

$$|U'|^2 - F'_\alpha \geq -\frac{1}{4} \sum_{i,j,k} W_{ij} W_{jk} W_{ki} (q_\alpha(s_{ij}) q_\alpha(s_{ki}) + q_\alpha[s_{ij}, s_{ki}]). \quad (2.11)$$

Observe that

$$W_{ij} W_{jk} W_{ki} = (e_i \cdot U)^2 (e_j \cdot U)^2 (e_k \cdot U)^2 \geq 0, \quad \sum_{i,j,k} W_{ij} W_{jk} W_{ki} = \text{Tr}(W^3) = |U|^6.$$

Thus, to prove the pointwise inequality in (2.8), it suffices to show that

$$q_\alpha(s) q_\alpha(t) + q_\alpha[s, t] \leq -\alpha^2 \quad \text{for all } s, t \in [0, 1]. \quad (2.12)$$

For $s \neq t$, set $X = \alpha(s - 1/2)$, $Y = \alpha(t - 1/2)$ and $z = X - Y$. The trigonometric addition formulas give

$$q_\alpha(s) q_\alpha(t) + q_\alpha[s, t] + \alpha^2 = \frac{\alpha^2}{\cos X \cos Y} \left(\cos z - \frac{\sin z}{z} \right). \quad (2.13)$$

Here $\cos X \cos Y > 0$ and $|z| \leq \alpha < \pi$. Since

$$\frac{d}{dz} (\sin z - z \cos z) = z \sin z \geq 0 \quad (0 \leq z < \pi),$$

we have $z \cos z \leq \sin z$ for $0 < z < \pi$. By evenness, the right-hand side of (2.13) is non-positive. The identity extends to the diagonal by continuity, proving (2.12). Since every s_{ij} belongs to $[0, 1]$, inserting (2.12) into (2.11) gives

$$|U'|^2 - F'_\alpha \geq \frac{\alpha^2}{4} |U|^6.$$

Integrating the above over $(-\Lambda, \Lambda)$ and using the local absolute continuity of F_α , we obtain

$$\int_{-\Lambda}^{\Lambda} |U'|^2 dx - F_\alpha(\Lambda) + F_\alpha(-\Lambda) \geq \frac{\alpha^2}{4} \int_{-\Lambda}^{\Lambda} |U|^6 dx.$$

To show that $F_\alpha(\Lambda)$ and $F_\alpha(-\Lambda)$ go to zero as $\Lambda \rightarrow \infty$, we use

$$|F_\alpha| \leq \frac{1}{4} \|q_\alpha\|_{L^\infty(0,1)} \|W\|_{\text{HS}}^2 = \frac{1}{4} \|q_\alpha\|_{L^\infty(0,1)} |U|^4.$$

Since $U \in H^1(\mathbb{R}; \mathbb{R}^N)$, it is bounded, and $(|U|^2)' = 2U \cdot U' \in L^1$ gives limits of $|U|^2$ at both ends, necessarily zero because $|U|^2 \in L^1$. Thus $F_\alpha(\pm\infty) = 0$, and both $|U|^6$ and $|U'|^2$ are integrable. Letting $\Lambda \rightarrow \infty$ and then $\alpha \uparrow \pi$ gives (2.1),

$$\int_{\mathbb{R}} |U'|^2 dx \geq \frac{\pi^2}{4} \int_{\mathbb{R}} |U|^6 dx. \quad \square$$

Remark 6. Proposition 3 admits a matrix analogue of the Benguria–Loss Poincaré formulation. Put $S = \mathcal{M}_A$ and choose a symmetric matrix H satisfying $HU = 2U'$ almost everywhere, setting $H = 0$ where $U = 0$. Then

$$W' = S'H, \quad S' = \mathcal{M}_W, \quad S(-\infty) = 0, \quad S(+\infty) = I.$$

Thus $dW = (dS)H$ replaces the scalar relation $dw = \dot{w} ds$. Moreover,

$$\langle H, S'H \rangle_{\text{HS}} = 4|U'|^2, \quad \langle W, S'W \rangle_{\text{HS}} = |U|^6.$$

Consequently, the proposition is equivalent to

$$\int_{\mathbb{R}} \langle H, (dS)H \rangle_{\text{HS}} \geq \pi^2 \int_{\mathbb{R}} \langle W, (dS)W \rangle_{\text{HS}}, \quad W(\pm\infty) = 0,$$

where integration against dS means integration against $S'(x) dx$. For $N = 1$, $S = a$ and $H = \dot{w}$, so this is precisely the Dirichlet Poincaré inequality on $(0, 1)$. The matrix formulation uses the compatibility $W = UU^\top$ and $dS = \mathcal{M}_W dx$.

3. PROOF OF THEOREM 2

We now extend the one-dimensional Lieb–Thirring inequality to vector-valued orthonormal families, retaining the sharp constant. As in Section 2, we state the equivalent kinetic formulation:

Proposition 7. *Let N, m be positive integers and let $u_1, \dots, u_N \in H^1(\mathbb{R}; \mathbb{R}^m)$ be orthonormal in $L^2(\mathbb{R}; \mathbb{R}^m)$. Define the non-negative density matrix*

$$R(x) = \sum_{j=1}^N u_j(x) u_j(x)^\top.$$

Then

$$\sum_{j=1}^N \int_{\mathbb{R}} |u_j'(x)|^2 dx \geq \frac{\pi^2}{4} \int_{\mathbb{R}} \text{Tr } R(x)^3 dx.$$

The proof of Proposition 7 is given in Subsection 3.1. We first derive Theorem 2 by the standard lifting argument [LW00, DLL08].

Proof of Theorem 2. For $d = 1$, the assertion is Theorem 1. By the standard duality [DLL08, Section 3] and finite-dimensional approximation as in [LW00, Section 2.3], Proposition 7 gives the operator-valued Lieb–Thirring inequality

$$\text{Tr} \left(-\frac{d^2}{dx^2} \otimes I_{\mathcal{H}} - \mathcal{V} \right)_- \leq \frac{4}{3\sqrt{3}\pi} \int_{\mathbb{R}} \text{Tr}_{\mathcal{H}} \mathcal{V}(x)^{3/2} dx, \quad (3.1)$$

for strongly measurable non-negative compact operator-valued potentials $\mathcal{V}$ on a separable real Hilbert space $\mathcal{H}$ with $\int_{\mathbb{R}} \operatorname{Tr}_{\mathcal{H}} \mathcal{V}(x)^{3/2} dx < \infty$.

For $d \geq 2$, take $\mathcal{H} = L^2(\mathbb{R}^{d-1}; \mathbb{R})$ and $\mathcal{V}(x_1) = (-\Delta_{d-1} - V_+(x_1, \cdot))_-$. The standard lifting argument [LW00] combines (3.1) with the sharp $\gamma = 3/2$ inequality in the remaining coordinates and gives

$$L_{1,d} \leq \frac{4}{3\sqrt{3}\pi} L_{3/2,d-1}^{\text{cl}} = \frac{2}{\sqrt{3}} L_{1,d}^{\text{cl}}. \quad \square$$

3.1. The vector-valued extension. In this subsection we prove Proposition 7. As in Section 2, the scalar case, we want a bound for the remainder coming from the completion of the square argument. We state this as an auxiliary lemma below.

Lemma 8. *Let $\alpha \in (0, \pi)$, let P be a real symmetric non-negative $N \times N$ matrix and let $\mu \in [0, 1]^N$. For $s_{ij} = (\mu_i + \mu_j)/2$, we have*

$$\mathcal{G}_{\alpha}(P, \mu) = - \sum_{i,j,k=1}^N P_{ij} P_{jk} P_{ki} [\alpha^2 + q_{\alpha}(s_{ij}) q_{\alpha}(s_{kj}) + q_{\alpha}[s_{ij}, s_{kj}]] \geq 0.$$

Recall that q_{α} was already fixed in (2.6) and that $q_{\alpha}[s, t] = (q_{\alpha}(s) - q_{\alpha}(t))/(s - t)$, with $q_{\alpha}[s, s] = \dot{q}_{\alpha}(s)$. The proof of Lemma 8 is the content of Subsection 3.2. Assuming it for now, we give the following:

Proof of Proposition 7. Let $U \in \mathbb{R}^{N \times m}$ have rows $u_j^{\top}$ and set

$$W = UU^{\top}, \quad A(x) = \int_{-\infty}^x W(y) dy, \quad R = U^{\top} U.$$

By orthonormality,

$$A' = W, \quad A(-\infty) = 0, \quad A(+\infty) = I, \quad 0 \leq A \leq I.$$

Fix $\alpha \in (0, \pi)$ and take the same correction as in (2.9),

$$F_{\alpha} = \frac{1}{4} \langle W, q_{\alpha}(\mathcal{M}_A) W \rangle_{\text{HS}}.$$

At a fixed point, let μ_i be the eigenvalues of A and set $s_{ij} = (\mu_i + \mu_j)/2$. Express W in an orthonormal eigenbasis of A , and write $B = q_{\alpha}(\mathcal{M}_A) W$. Completing the square as in (2.10), the same calculation from Section 2, with indices relabelled, gives

$$\begin{aligned} \|U'\|_{\text{HS}}^2 - F'_{\alpha} - \frac{\alpha^2}{4} \operatorname{Tr} W^3 &= \left\| U' - \frac{1}{2} B U \right\|_{\text{HS}}^2 \\ &\quad - \frac{1}{4} \sum_{i,j,k=1}^N W_{ij} W_{jk} W_{ki} [\alpha^2 + q_{\alpha}(s_{ij}) q_{\alpha}(s_{kj}) + q_{\alpha}[s_{ij}, s_{kj}]] \geq 0. \end{aligned} \quad (3.2)$$

The differentiation uses $W' = U'U^{\top} + U(U')^{\top}$ and does not require rank one. Lemma 5 justifies it almost everywhere and gives local absolute continuity of F_{α} ; the cubic remainder is $\mathcal{G}_{\alpha}(W, \mu)/4$, which is non-negative by Lemma 8.

As in Section 2, U is bounded and tends to zero at both ends of the line, and

$$|F_\alpha| \leq \frac{1}{4} \|q_\alpha\|_\infty \|W\|_{\text{HS}}^2 \leq \frac{1}{4} \|q_\alpha\|_\infty \|U\|_{\text{HS}}^4.$$

Integrating (3.2) and letting $\alpha \uparrow \pi$ proves the assertion, since $\text{Tr } W^3 = \text{Tr } R^3$ and $\|U'\|_{\text{HS}}^2 = \sum_j |u'_j|^2$. $\square$

3.2. Lower bound on the remainder. For convenience let us introduce the auxiliary functions

$$h(t) = \frac{\sin t}{t} - \cos t \quad \text{and} \quad g(x, y) = \cos(x + y)h(x - y).$$

Equipped with these functions we give a proof of Lemma 8.

Proof of Lemma 8. Fix α , P and μ as in the statement, and put

$$\theta_i = \frac{\alpha}{2} \left(\mu_i - \frac{1}{2} \right), \quad C_{ij} = \frac{P_{ij}}{\cos(\theta_i + \theta_j)}.$$

By (2.13),

$$\alpha^2 + q_\alpha(s_{ij})q_\alpha(s_{kj}) + q_\alpha[s_{ij}, s_{kj}] = -\frac{\alpha^2 h(\theta_i - \theta_k)}{\cos(\theta_i + \theta_j) \cos(\theta_k + \theta_j)},$$

and hence

$$\mathcal{G}_\alpha(P, \mu) = \alpha^2 \sum_{i,k} (C^2)_{ik} C_{ki} g(\theta_i, \theta_k). \quad (3.3)$$

The matrix C is non-negative. Indeed, let $D = \text{diag}((\cos \theta_i)^{-1})$ and $T = \text{diag}(\tan \theta_i)$. The cosine addition formula gives

$$C - TCT = DPD \geq 0.$$

If ν is the smallest eigenvalue of C and v is a corresponding unit eigenvector, then $C \geq \nu I$ and

$$\nu = \langle v, Cv \rangle \geq \langle Tv, CTv \rangle \geq \nu \|Tv\|^2.$$

Since $|\theta_i| < \pi/4$, we have $\|T\| < 1$, and therefore $\nu \geq 0$.

For real numbers $\sigma_1, \dots, \sigma_N$, set $K_{ik} = (\sigma_i - \sigma_k)^2 C_{ik}$. We claim that

$$\sum_{i,k} (C^2)_{ik} C_{ki} (\sigma_i - \sigma_k)^4 = 2 \text{Tr}(CK^2) = 2 \|C^{1/2} K\|_{\text{HS}}^2 \geq 0. \quad (3.4)$$

To verify the first equality, write

$$\text{Tr}(CK^2) = \sum_{i,j,k} C_{ij} C_{jk} C_{ki} (\sigma_j - \sigma_k)^2 (\sigma_k - \sigma_i)^2.$$

For $a = \sigma_i - \sigma_j$, $b = \sigma_j - \sigma_k$ and $c = \sigma_k - \sigma_i$, the relation $a + b + c = 0$ implies

$$a^4 + b^4 + c^4 = 2(a^2 b^2 + b^2 c^2 + c^2 a^2).$$

Multiply by $C_{ij} C_{jk} C_{ki}$ and sum over i, j, k . Cyclic relabelling makes the three sums on each side equal, giving (3.4). The second equality follows from $K^\top = K$ and $C \geq 0$.

Assume for the moment that $g(\theta_i, \theta_k) = |v_i - v_k|^4$ for some vectors $v_1, \dots, v_N \in \mathbb{R}^m$. We prove their existence in Lemma 9 at the end of this section. For $z \in \mathbb{R}^m$, rotational invariance gives

$$\int_{\mathbb{S}^{m-1}} (\omega \cdot z)^4 d\omega = c_m |z|^4, \quad c_m = \frac{3|\mathbb{S}^{m-1}|}{m(m+2)}.$$

Here $d\omega$ is surface measure; when $m = 1$, the integral is the sum over $\{-1, 1\}$ and $c_1 = 2$. Put

$$K(\omega)_{ik} = (\omega \cdot v_i - \omega \cdot v_k)^2 C_{ik}.$$

Equations (3.3) and (3.4) now give

$$\mathcal{G}_\alpha(P, \mu) = \frac{2\alpha^2}{c_m} \int_{\mathbb{S}^{m-1}} \text{Tr}(CK(\omega)^2) d\omega \geq 0. \quad \square$$

Lemma 9. *Let $x_1, \dots, x_N \in (-\pi/4, \pi/4)$. Then there exist an integer $m \geq 1$ and vectors $v_1, \dots, v_N \in \mathbb{R}^m$ such that*

$$g(x_i, x_j) = |v_i - v_j|^4, \quad 1 \leq i, j \leq N.$$

Proof of Lemma 9. Set $r(x, y) = \sqrt{g(x, y)}$. The function h is even and

$$h(t) = \frac{1}{t} \int_0^t s \sin s ds > 0, \quad 0 < t \leq \pi/2,$$

so r is well defined on $(-\pi/4, \pi/4)^2$. Put

$$\mathcal{D}(x) = \frac{2\sqrt{\cos(2x)}}{\sqrt{3}}, \quad \mathcal{K}(x, y) = -\partial_x \partial_y r(x, y) \quad (x \neq y).$$

Since $h(t)/t^2$ extends to a smooth even function with value $1/3$ at zero, $\mathcal{K}$ extends continuously across the diagonal with $\mathcal{K}(x, x) = 0$, and

$$\lim_{y \uparrow x} \partial_x r(x, y) = \frac{\mathcal{D}(x)}{2}.$$

For $y < x$, integration in the second variable gives

$$\partial_x r(x, y) = \frac{\mathcal{D}(x)}{2} + \int_y^x \mathcal{K}(x, t) dt.$$

Integrating in the first variable, using $r(y, y) = 0$ and the symmetry of $\mathcal{K}$, yields

$$r(x, y) = \frac{1}{2} \int_y^x \mathcal{D}(s) ds + \frac{1}{2} \int_y^x \int_y^x \mathcal{K}(s, t) dt ds. \quad (3.5)$$

On the real vector space of compactly supported step functions on $(-\pi/4, \pi/4)$, identified up to equality almost everywhere, define

$$\mathcal{B}(f_1, f_2) = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \mathcal{D}(s) f_1(s) f_2(s) ds + \frac{1}{2} \iint_{(-\pi/4, \pi/4)^2} \mathcal{K}(s, t) f_1(s) f_2(t) ds dt.$$

We first prove that this symmetric bilinear form is positive definite.

Integration by parts gives, for $0 < t \leq \pi/2$,

$$0 < h(t) = \frac{t^2}{2} \int_0^1 (1 - u^2) \cos(tu) \, du \leq \frac{t^2}{3},$$

$$0 \leq \frac{2}{t} - \frac{h'(t)}{h(t)} = \frac{t \int_0^1 (1 - u^2)^2 \cos(tu) \, du}{4 \int_0^1 (1 - u^2) \cos(tu) \, du} \leq \frac{t}{4}.$$

Here $0 \leq 1 - u^2 \leq 1$ and $\cos(tu) \geq 0$. Thus $0 < h'/h \leq 2/t$ and

$$0 \leq \frac{4}{t^2} - \left(\frac{h'(t)}{h(t)} \right)^2 = \left(\frac{2}{t} - \frac{h'(t)}{h(t)} \right) \left(\frac{2}{t} + \frac{h'(t)}{h(t)} \right) \leq 1.$$

Using $h'' = (2/t^2 - 1)h$ and writing $t = |x - y| > 0$, differentiation gives

$$\mathcal{K}(x, y) = \frac{r(x, y)}{4} \left[\tan^2(x + y) + \frac{4}{t^2} - \left(\frac{h'(t)}{h(t)} \right)^2 \right].$$

Consequently,

$$0 \leq \mathcal{K}(x, y) \leq \frac{t}{4\sqrt{3} \cos^{3/2}(x + y)} \leq \frac{\pi}{8\sqrt{3}} \frac{|\sin(x - y)|}{\cos^{3/2}(x + y)}.$$

The last inequality uses $t \leq (\pi/2) \sin t$ for $0 \leq t \leq \pi/2$; the bounds also hold on the diagonal.

For fixed $x \in (-\pi/4, \pi/4)$, put

$$w(x) = \cos(2x)^{-3/4}, \quad J_x(y) = \frac{\cos^{1/4}(2y)}{\sqrt{\cos(x + y)}}.$$

Differentiation yields

$$J'_x(y) = \frac{w(y) \sin(x - y)}{2 \cos^{3/2}(x + y)}, \quad \mathcal{K}(x, y)w(y) \leq \frac{\pi}{4\sqrt{3}} |J'_x(y)|.$$

The function J_x vanishes at both endpoints, increases for $y < x$ and decreases for $y > x$. Hence

$$\int_{-\pi/4}^{\pi/4} |J'_x(y)| \, dy = 2J_x(x) = 2 \cos(2x)^{-1/4},$$

where the integrals are limits over interior intervals. We conclude that

$$\int_{-\pi/4}^{\pi/4} \mathcal{K}(x, y)w(y) \, dy \leq \frac{\pi}{2\sqrt{3}} \cos(2x)^{-1/4} = \frac{\pi}{4} \mathcal{D}(x)w(x). \quad (3.6)$$

By symmetry of $\mathcal{K}$ and the inequality

$$2|f(x)f(y)| \leq f(x)^2 \frac{w(y)}{w(x)} + f(y)^2 \frac{w(x)}{w(y)},$$

the estimate (3.6) implies

$$\begin{aligned} \left| \iint_{(-\pi/4, \pi/4)^2} \mathcal{K}(x, y) f(x) f(y) \, dx \, dy \right| &\leq \int_{-\pi/4}^{\pi/4} \frac{f(x)^2}{w(x)} \left(\int_{-\pi/4}^{\pi/4} \mathcal{K}(x, y) w(y) \, dy \right) dx \\ &\leq \frac{\pi}{4} \int_{-\pi/4}^{\pi/4} \mathcal{D}(x) f(x)^2 \, dx. \end{aligned}$$

Therefore

$$\mathcal{B}(f, f) \geq \frac{1}{2} \left(1 - \frac{\pi}{4} \right) \int_{-\pi/4}^{\pi/4} \mathcal{D}(x) f(x)^2 \, dx.$$

Since $\mathcal{D} > 0$ and $\pi/4 < 1$, the form $\mathcal{B}$ is positive definite.

Finally, let $x_* = \min_i x_i$ and $\chi_i = \mathbf{1}_{(x_*, x_i]}$. Equation (3.5) and symmetry give

$$r(x_i, x_j) = \mathcal{B}(\chi_i - \chi_j, \chi_i - \chi_j).$$

If $E = \text{span}\{\chi_1, \dots, \chi_N\} = \{0\}$, all the x_i coincide, and we take $m = 1$ and $v_i = 0$. Otherwise, set $m = \dim E$, choose a $\mathcal{B}$ -orthonormal basis $e_1, \dots, e_m$, and define $v_i \in \mathbb{R}^m$ by

$$\chi_i = \sum_{\ell=1}^m v_{i\ell} e_\ell, \quad v_i = (v_{i1}, \dots, v_{im}).$$

Then $|v_i - v_j|^2 = \mathcal{B}(\chi_i - \chi_j, \chi_i - \chi_j) = r(x_i, x_j)$, and squaring proves the assertion. $\square$

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Use of AI. A first version of the main argument was developed by GPT-6 Astra under the guidance of L.R., following collaboration over a couple of months with successive models, beginning with GPT-5.5. Extensions and improvements were then jointly obtained by M.R.S. and L.R. using Fable 5.1 and GPT-6, respectively. The argument was further simplified with assistance from the models, then restructured and verified by the authors. The authors take full responsibility for the final version of the proof.

We address the timeline of how the argument was made public. Our first version of the argument was uploaded to arXiv on the 9th of September 2026. On the 6th of October 2026, OpenAI released a batch of results, including results in mathematical physics, on GitHub. One such result, dated the 23rd of September, addresses the Lieb–Thirring inequality in one dimension for the full range $1/2 < \gamma < 3/2$. It builds on our argument by adding an additional optimisation problem. Our work may have been a substantial input for them. They clearly acknowledge our contribution in establishing the case $\gamma = 1$.

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